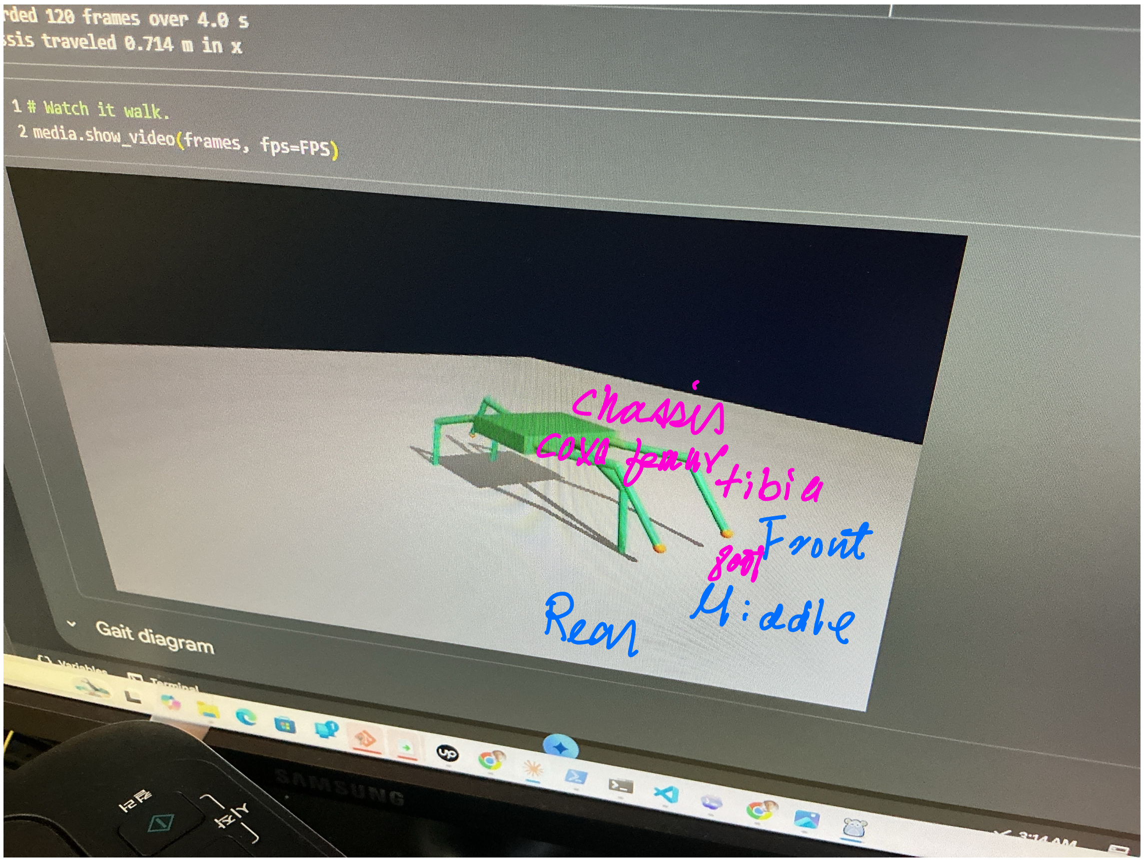

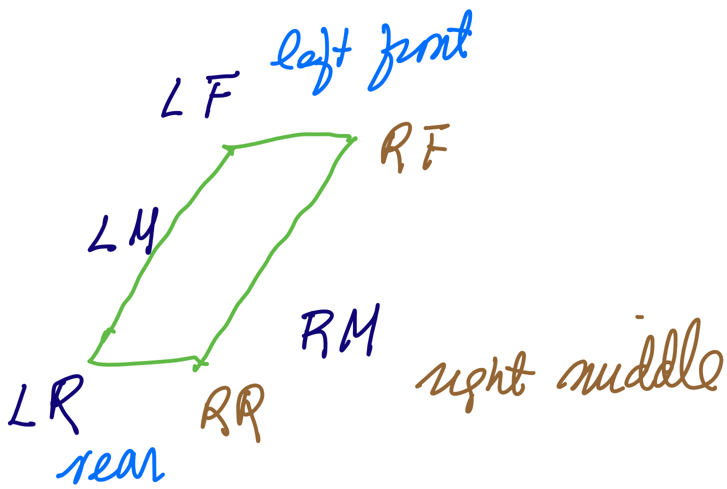

Central Pattern Generator

pod A: LF, RM, LR — phase offset 0

pod B: RF, LM, RR — phase offset 0.5

al opposites move together; at any moment three legs (one front + one mid + one

```python
# Tripod assignment (group 0 = tripod A, group 1 = tripod B).
TRIPOD_GROUP = {"LF": 0, "RM": 0, "LR": 0, "RF": 1, "LM": 1, "RR": 1}

def cpg_joints(t, leg_name,
               period=0.7, coxa_amp=22.0,
               femur_base=-20.0, femur_lift=30.0,
               tibia_base=-60.0):
    """Return (coxa, femur, tibia) target angles in DEGREES for one leg at tim
    offset = 0.5 if TRIPOD_GROUP[leg_name] == 1 else 0.0
    phase = ((t / period) + offset) % 1.0

    # Coxa: cosine swing.  +amp at phase 0 (forward), -amp at phase 0.5 (back).
    coxa = coxa_amp * np.cos(2 * np.pi * phase)

    # Femur: stay at base during stance, lift in an arc during swing.
    if phase < 0.5:
        femur = femur_base                          # stance
    else:
        s = (phase - 0.5) / 0.5                     # swing
        femur = femur_base + femur_lift * np.sin(np.pi * s)

    # Tibia: keep the bend constant for simpl
    tibia = tibia_base
    return coxa
```

Commands + Code + Text Run all

You said

The mid

fore-aft

+0.16),

So the th

Leg

LF

LM

LR

with spa

Putting

Here's th

```python
def make_leg(name, attach_x, side):
    """Return the XML for one leg attached to the chassis.

    Convention: positive coxa  = leg swings toward +x (forward)
                positive femur = leg lifts up (foot toward +z)
                positive tibia = knee bends (foot tucks under)
    Left/right symmetry is achieved by flipping the axis signs on the right side.
    """
    if side == 'L':
        attach_y = +CHASSIS_LY
        leg_y_sign = +1
        coxa_axis  = "0 0 -1"                    # leg extends in +y
        lift_axis  = "1 0 0"                     # +coxa = forward (left side)
                                                 # +femur = lift up (left side)
    else:
        attach_y = -CHASSIS_LY
        leg_y_sign = -1
        coxa_axis  = "0 0 1"                     # leg extends in -y
        lift_axis  = "-1 0 0"                    # +coxa = forward (right side)
                                                 # +femur = lift up (right side)
    fy = leg_y_sign
    return f"""
      <body name=\"{name}_coxa\" pos=\"{attach_x} {attach_y} 0\">
        <joint name=\"{name}_coxa_j\"  axis=\"{coxa_axis}\" range=\"-45 45\"/>
        <geom  name=\"{name}_coxa_g\"  type=\"capsule\" size=\"0.012\"
               fromto=\"0 0 0  0 {fy*COXA_LEN} 0\" rgba=\"0.30 0.50 0.30 1\"/>
        <body name=\"{name}_femur\" pos=\"0 {fy*COXA_LEN} 0\">
          <joint name=\"{name}_femur_j\" axis=\"{lift_axis}\" range=\"-60 60\"/>
          <geom  name=\"{name}_femur_g\" type=\"capsule\" size=\"0.011\"
                 fromto=\"0 0 0  0 {fy*FEMUR_LEN} 0\" rgba=\"0.20 0.65 0.40 1\"/>
          <body name=\"{name}_tibia\" pos=\"0 {fy*FEMUR_LEN} 0\">
            <joint name=\"{name}_tibia_j\" axis=\"{lift_axis}\" range=\"-30 90\"/>
            <geom  name=\"{name}_tibia_g\" type=\"capsule\" size=\"0.009\"
                   fromto=\"0 0 0  0 {fy*TIBIA_LEN} 0\" rgba=\"0.10 0.75 0.45 1\"/>
            <site name=\"{name}_foot\" pos=\"0 {fy*TIBIA_LEN} 0\"
                  size=\"0.012\" rgba=\"1 0.4 0 1\"/>
          </body>
        </body>
```

coxa

femur

tibia

foot

coxa

femur

tibia

foot